Boudrouss Réda | CV

CIFRE PhD candidate – Static analysis & abstract interpretation – Paris

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Languages: French (Native), English (Professional), Arabic (Conversational)

Profile

Completing an MSc in Software Science and Technology at Sorbonne University, I am looking for a **partner company for an industrial PhD (CIFRE)** in **static analysis and abstract interpretation**, supervised by **Antoine Miné** (LIP6). I am particularly interested in the analysis of **WebAssembly**, **Rust**, and dynamic languages such as **JavaScript**, as well as **OCaml**. I ported the **MOPSA** analyzer (OCaml/C/C++) to WebAssembly and developed an abstract-interpretation-based analyzer in Rust. My apprenticeship at Hexaglobe gave me hands-on experience in **software verification in an industrial setting**.

Education

MSc in Software Science and Technology - Sorbonne University 2024 – 2026

· Coursework: **abstract interpretation** and type systems (course led by Antoine Miné), inference of numerical properties, **formal program verification**, static typing, design and implementation of **programming languages** and their runtimes (interpreters, compilers), **concurrent** and distributed programming.

BSc in Computer Science - Sorbonne University 2020 – 2024

· Algorithms, computer architecture, operating systems, networks, databases and functional programming.

Projects

Mopsa-wasm - github.com/rboudrouss/mopsa-emcc 2026

· Ported **MOPSA**, an **abstract-interpretation** static analyzer for C and Python written in **OCaml**/C/C++ at LIP6, to **WebAssembly**, compiling the OCaml runtime and the full native stack (Clang frontend, **Apron**, GMP, MPFR) into a single static Wasm module with Emscripten.

· Worked on both sides of the **OCaml runtime / C FFI** boundary, generating a static table of ∼1,400 C primitives (no `dlopen` on Wasm) and fixing an unsigned-wraparound bug in the runtime's `Tag_val` macro. Recompiled Apron's numerical domains to use exact GMP rationals, addressing the **soundness** hazard of Wasm's fixed rounding mode.

· Shipped a fully client-side playground and a technical write-up.

Reactant analyzer - github.com/rboudrouss/reactant-analyzer 2026

· Designed and implemented an **abstract-interpretation**-based static analyzer in **Rust** (∼30k lines, 500+ tests) that catches **React** hook bugs beyond the reach of syntactic linters, such as infinite render loops, derived state and unstable dependencies.

· Evaluates components in abstract domains (intervals, constant sets, reference-stability tags) over a dedicated CFG-based IR and flags an infinite render loop when the state fixpoint **widens**. Concrete semantics based on the React-tRace paper (OOPSLA 2025).

Experience

Quality Analyst Engineer (apprenticeship) - Hexaglobe (video streaming) 2025 – 2026

· Designed complete **test strategies** (unit, integration, E2E) for zero-coverage legacy products, reaching ∼**3,000 test cases** across 4 stacks (TypeScript, Python, PHP, JS).

· Built objective **quality oracles** (VMAF, black-frame/freeze detection, audio analysis) to automatically validate a live video encoder.

· Industrialized **AI-agent test production** gated by deterministic verification signals (sandboxed multi-agent orchestration in Docker, versioned guardrails, mandatory green tests, systematic human review), producing ∼720 integration tests in 3 days.

· Built **GitLab CI/CD** pipelines and shared CI components consumed by 10+ repositories, wiring **static analysis** tools into the chain.

Technical Lead, then President (2024–25) - Play Sorbonne Université Association 2023 – 2026

· Designed and operated a **virtualized infrastructure** (Docker servers, CI/CD, networking) and built a **local-first** mobile web app. As president, led **100+ volunteers** for a cultural event welcoming 10,000+ visitors.